

HASHMAP

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The biggest competition is myself. I am not looking to follow others or pull them down. I'm planning to test my own boundaries.

RAIN

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Good
Morning

Today's content = pair with given sum

⇒ count pair sum

⇒ count pair diff

⇒ Subarr sum = k

⇒ Distinct char in window = k

01. Given N array elements, check if there exists a pair (i, j) such that $arr[i] + arr[j] = k$ & $(i \neq j)$

arr [] :

8	9	1	-2	4	5	11	-6	7	5
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0123456789

K = 11 $\longrightarrow$ True (4 & 8 idx)

K = 6 $\longrightarrow$ True (0 & 3 idx)

K = 22 $\longrightarrow$ False

boolean HashSet sum (int[] arr, int sum) will contain ele from 0 to $i-1$

HashSet<Integer> hs = new HashSet<>();

for (int i = 0; i < n; i++) {

$x = arr[i]$, $y = sum - x$

if (hs.contains(y)) return true;

hs.add(x);

}

return false;

TC: $O(n^2)$

SC: $O(n)$

Count pair sum

Given an $arr[]$ & an integer B . Count no. of pairs (i, j) such that $A[i] + A[j] = B$ and $i \neq j$

Note:- The pair (i, j) is same as pair (j, i) & should be counted once.

$A = \{ \overset{0}{3}, \overset{1}{5}, \overset{2}{1}, \overset{3}{2} \}$ $B = 8$ Pair
 $Ans = 1$ $(0, 1)$

$A = \{ \overset{0}{1}, \overset{1}{2}, \overset{2}{1}, \overset{3}{2} \}$ $B = 3$ Pairs
 $Ans = 4$ $(0, 1) \quad (0, 3)$
 $(1, 2) \quad (2, 3)$

$arr[] :$

8	9	5	-2	4	5	5	9	6	5
---	---	---	----	---	---	---	---	---	---

Sum=10

↓

0 1 2 3 4 5 6 7 8 9

x	$y (sum - x)$	HM ↗ 0 to 9-1	ans
8	2	{ }	ans = 0
9	1	{ <8, 1> }	ans = 0
5	5	{ <8, 1> , <9, 1> }	ans = 0
-2	12	{ <8, 1> , <9, 1> , <5, 1> }	ans = 0
4	6	{ <8, 1> , <9, 1> , <5, 1> , <-2, 1> }	ans = 0

5	5	$\{ \langle 8,1 \rangle \langle 9,1 \rangle \langle 5,1 \rangle \langle -2,1 \rangle \}$ $\langle 4,1 \rangle$	ans = 1
5	5	$\{ \langle 8,1 \rangle \langle 9,1 \rangle \langle 5,2 \rangle \langle -2,1 \rangle \}$ $\langle 4,1 \rangle$	ans = 1 + 2 = 3
9	1	$\{ \langle 8,1 \rangle \langle 9,1 \rangle \langle 5,3 \rangle \langle -2,1 \rangle \}$ $\langle 4,1 \rangle$	ans = 3
6	4	$\{ \langle 8,1 \rangle \langle 9,2 \rangle \langle 5,3 \rangle \langle -2,1 \rangle \}$ $\langle 4,1 \rangle$	ans = 3 + 1 = 4
5	5	$\{ \langle 8,1 \rangle \langle 9,2 \rangle \langle 5,3 \rangle \langle -2,1 \rangle \}$ $\langle 4,1 \rangle \langle 6,1 \rangle$	ans = 4 + 3 ans = 7

HM < I, I > map = new HashMap<>();

ans = 0

for (i = 0; i < n; i++) {

 x = A[i] , y = B - A[i]

 if (map.containsKey(y)) {

 ans = ans + map.get(y)

 }

 map.put(x, map.getOrDefault(x, 0) + 1);

}

return ans;

Count pair diff

Given an array A & an integer B. Count the no. of pairs (i, j) such that $A[i] - A[j] = B$ and $i \neq j$

$$A = \{ \underset{0}{3}, \underset{1}{5}, \underset{2}{1}, \underset{3}{2} \}$$

$$B = 4$$

pair
 $(1, 2)$

Ans = 1

$$A = \{ \underset{0}{3}, \underset{1}{1}, \underset{2}{4}, \underset{3}{1}, \underset{4}{5} \}$$

$$B = 2$$

pair
 $(0, 1)$
 $(0, 3)$
 $(0, 4)$

Ans = 3

$$A = \{ \underset{0}{5}, \underset{1}{3}, \underset{2}{3}, \underset{3}{2}, \underset{4}{6} \}$$

$$B = 1$$

pairs $\Rightarrow (0, 4) (1, 3) (2, 3)$

Ans = 3

$$A = \{ \underset{0}{1}, \underset{1}{3}, \underset{2}{3}, \underset{3}{2} \}$$

$B = 1$

$2 - 1 = 1$
 $2 + 1 = 3$

HM

$$\left\{ \begin{array}{l} 1 \rightarrow 1 \\ 3 \rightarrow 2 \end{array} \right\}$$

$$\text{ans} = 1 + 2 \\ = \underline{3}$$

HM < I, I > map = new HashMap<>();

ans = 0

for (i = 0; i < n; i++) {

 x = A[i] , y = A[i] - B , z = B + A[i]

 if (map.containsKey(y)) {

 | ans = ans + map.get(y)

Tc: O(N)

Sc: O(N)

 if (map.containsKey(z)) {

 | ans = ans + map.get(z);

 map.put(x, map.getOrDefault(x, 0) + 1);

}
return ans;

A = { 3, 1, 4, 1, 5 } B = 2

 0 1 2 3 4

 ↓ S+B = 7

 S-B = 3

HM

{ 3 → 1
 1 → 2
 4 → 1
 5 → 1 }

ans = 1 + 1 = 2 + 1 = 3

8:11 AM → 8:21 AM

02. Subarr with given sum

Given an arr of positive integers A & an integer B. check if there exist a subarr which adds to K.

arr[] =

8	9	2	-2	4	5	11	-6
0	1	2	3	4	5	6	7

arr[] =

3	9	-2	4	1	-7	2	6	-5	8	-3	-7	6
0	1	2	3	4	5	6	7	8	9	10	11	12